

1 INTRODUCTION

The work involves programming the binary inputs and outputs as well as a temperature measurement application, which is visualized with the help of a pre-made control room application on the PC. Both temperature measurement applications and PC-based control rooms are very common in the industry. It is also typical that when the temperature exceeds a certain limit value, the system gives an alarm and performs any necessary safety functions to prevent damage.

The PC has a built-in WinCC control room application that exchanges data with programmable logic through the memory slots provided in this document. If WinCC is not available for some reason, the Watch Table from TIA Portal can be used for monitoring and testing (the instructor will advise you if necessary).

In the current temperature measurement application, a program is implemented that converts the INTEGER-type temperature data obtained from the memory word of the analogue input of logic into a REAL (floating point, i.e. enables decimal places), scales it to degrees Celsius, and transfers the scaled REAL-type temperature data later in this instruction to the specified memory location from where the WinCC control room application can read it.

After scaling, circuits are made in the program comparing the temperature with the temperature alarm limit entered from the WinCC control room (or Watch table). If the temperature is higher than the given alarm limit, the output of the programmable logic must be turned on. At the same time, the marker-bit is also set to the mode '1' and an alarm light will light up in the WinCC control room. All input, output, and memory addresses will be provided later in this document.

Before the temperature measurement application, circuits are performed as a separate exercise in the work, in which the modes of the simulation switches installed on the

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inputs of logic are read. Based on the status information, actions are performed according to the truth table that will be presented later.

2 INFORMATION ABOUT THE WORKSTATION**S7 1214C DC/DC/DC**

The logic of the S7 1200 series is used in the work. Check the number and locations of the binary inputs, outputs and analogue inputs in the device. If necessary, ask the instructor for advice.

Temperature transmitter

The work uses a temperature transmitter for building automation purposes, which uses 0... 10V voltage signal. The resistance of the RTD connected to the transmitter changes as the temperature changes, and the transmitter part of the device "converts" the change in sensor resistance to such an extent that it causes a standard voltage signal of 0..10 V. Check the tuning range of the transmitter you are using together with the instructor.

According to the 0...10 V voltage signal the analog input of programmable logic (the one to which the transmitter is connected) converts the analog voltage message into a digital format that can be read into the program as an integer (0... 27648).

Analog measurements and standard messages

In industry, the measuring current circuit connected to the measurement or automation system includes the following main parts: sensor, transmitter (or amplifier) and input module (or input card).

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The most common analog signal in industrial applications is 4... 20 mA. Other commonly used standard messages include 0... 10V, $\pm 10V$ and 0... 20 mA. The magnitude of the standard message at any given point in time depends on the value to be measured and the excitation range of the transmitter/amplifier used.

Example: If 4... 20 mA transmitter/amplifier is tuned to a temperature range of 0... 100 °C, and the temperature to be measured is 50 °C, is the magnitude of the current message passing through the measuring circuit:

$$\frac{20mA - 4mA}{100^{\circ}C} * 50^{\circ}C + 4mA = 12mA$$

Similarly: if 4...20 mA transmitter/amplifier is tuned to a temperature range of 0... 100 °C and the magnitude of the current signal in the measuring circuit of 10 mA is the magnitude of the measured temperature:

$$\frac{100^{\circ}C}{20mA - 4mA} * (10mA - 4mA) = 37,5^{\circ}C$$

In addition to standard analog messages, digital messages are used. In this case, the measuring device is connected to the control system in the so-called control system. on the fieldbus (eg. Profibus) and the data transfer has been implemented in serial form in digital format.

WinCC

WinCC is a so-called control room software, which can be used to create an interface between the person who is monitoring or supervising the process and the automation system (e.g. programmable logic and the devices connected to it). Through the user interface, it is possible, for example, to assign different setpoints to an application run

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with programmable logic (in this task, the temperature alarm limit) or to display some process quantities and events to a person in an illustrative form (in this task, a temperature indication and an alarm light indicating excess heat). Often, the control room application also involves data collection, which means that certain measurements and other key variables are stored in a database, from where they can be viewed later, for example, in connection with the investigation of fault situations.

The connection between the WinCC control room and the device controlling the process can be implemented, for example, via ethernet or MPI interface. The MPI (Multi Point Interface) interface is a commonly used method in Siemens' older S7 300 and 400 series logics to connect programmable logic to a programming device (i.e. PC), from which an application made with Step7 is loaded into programmable logic. The current S7 1200 and 1500 series logics have an integrated ethernet interface through which the program can be transferred.

3 SAFETY

At the workstation, normal care should be taken when working with electrical appliances. If there are any problems while working, ask the staff for advice. When you stop working, take any tools in place, clean the workstation and turn off the equipment.

N.B! Do not make changes to the measurement coupling without permission/instructions, as an incorrectly made measurement coupling may break the analogue input of programmable logic.

TEMPERATURE MEASUREMENT SIEMENS S7 1200**4 TASK**

1. Familiarize yourself with the necessary equipment and study how the temperature transmitter is connected to the programmable logic. Where does the transmitter get its operating voltage? How to export the temperature message to logic? To which connectors of logic is the transmitter connected? What is the range of the temperature transmitter?
2. Create a TIA Portal project and select the same CPU as the logic that is in the workcase. In the logic settings, under AI, check the input channel address (IW64 by default). Already at this stage, test the loading of the configuration and an empty program into the logic in order to avoid later, for example, loading the configuration into the logic. problems caused by incorrect device revision.
3. Make a function (FC1) for an application related to simulation switches.
4. Make an application in FC1 that reads the simulation switch states and implements the truth table in Appendix 2, while taking into account the "Enable/Disable" switch in the WinCC control room (also look at Appendix 1, which indicates the addresses to be used). If the switch is in the " Enable" position (in which case M90.0 = '1'), the outputs will operate under the conditions specified in the truth table. If the switch is in the "Disable" position (in which case M90.0 = '0') FC1 shall not direct any output or marker to '1'.
5. Save the finished FC1. Be sure to add subroutine call to the main program. Load the configuration and application program you made into the logic. Then try FC1 in monitoring mode. After trying FC1, close it. N.B! Since the value of M90.0 comes from the wincc control room (which has not yet been started at this stage of the exercise), there is little more that can be done in this intermediate testing than to detect a change in the modes of the simulation switches.

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6. Make another function (FC2) in the project for which you will make a temperature measurement application. Comment on the circuits of the function and give the addresses rational symbolic names. The addresses to be used are given in Annex 1. Make sure that the data types for the specified variables are correct. In the circuits of function 2, do the following:

- Convert the transferred (16-bit) INTEGER-type value (to 32-bit) REAL-type value and save it to a free memory slot (e.g. MD52). (In the Converter folder you can find the data type conversions)
- Scale the REAL-type temperature value to Celsius and save the result in the MD70 memory slot. N.B! the address must be specifically MD70, as the control room application assumes that the temperature information can be found at that memory address. If necessary, review the scaling from Moodle's lecture materials (Analog signal).
- Make a comparison between the values in memory slot MD70 (=measured temperature) and MD74 (=overheat alarm limit given from WinCC). If the temperature information in memory slot MD70 is greater than the alarm limit in memory slot MD74, set the marker M82.1 and output Q0.0 on. M82.1 is a marker bit from which the WinCC control room "sees" whether the temperature alarm limit has been exceeded. Q0.0 is the first physical output of logic.

7. Save the finished FC2. Be sure to call the subroutine in the main program and load the application into logic. In monitoring mode, test that the temperature value is displayed in the memory slot MD70. Please note that since the WinCC control room has not yet been started, the application does not receive any temperature alarm limit information from anywhere. Therefore, the alarm limit is likely to be 0, which results in the temperature exceeding the alarm limit, causing an alarm according to the application (Q0.0 = '1', etc.). After testing the application, and saving it, close FC2, leave the logic in RUN mode and go to the next step (WinCC control room experiment). If

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WinCC is not available, discuss with your instructor how to use Watch Table to test your program.

8. Once both functions are done, tested and transferred to logic: Next, start the WinCC control room from the desktop. If the control room goes directly into the so-called runtime mode, you can test your application with it. If for some reason the runtime does not start automatically, do the following: On the left side of the Control center (or rather WinCCExplorer) there is a section Graphics Designer, just click on it. Now, on the right side of the screen, you should open a section with the text "lampotila.pdl" (= control room display file) or some similarly named screen. Double-click on the text and the control room screen should open in Graphics Editor. Once the screen has opened, select the file runtime to turn on the control room app and test your app → Runtime, in which case the control room application will turn on and you will be able to test your application.

9. After testing the program you have made so far in the WinCC control room, make a warning circuit(s) in FC2 to get a warning that the temperature is close to the actual alarm limit. Warning circuits should place the marker M82.0 on the head (causing the yellow warning light to light up in the WinCC control room) when the temperature is no more than 4.0 degrees from the alarm limit. However, if the temperature rises above the actual alarm limit, the yellow light (M82.0) must go out, in which case the red alarm light will light up on its own.

10. When the work is completed, present it to the teacher (indicate the activity, show the program you have made, show the symbol table).

5 ATTACHMENTS

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Annex 1. I/O's addresses

ADDRESS	Data type	EXPLANATION
IW64	WORD	The address of the input area from which the temperature obtained from the measuring device is read into the logic. Used in FC2.
Q0.0	BOOL	The address of the output, which turns on the LED that notifies the alarm (in real life, a warning light, a buzzer, etc.). Used in FC2.
MD70	REAL	A memory slot where the WinCC control room reads the temperature. Used in FC2.
MD74	REAL	The memory slot where the WinCC control room writes the alarm limit for the given temperature. Used in FC2.
M82.1	BOOL	Alarm signal. The bit by which the WinCC control room turns on a red indicator when the temperature rises above the given alarm limit. Used in FC2.
M82.0	BOOL	Warning sign. The bit by which the WinCC control room lights up a yellow indicator light for information that the temperature is close to the alarm limit.
I0.0 - 0.7	BOOL	Input addresses of simulation switches. ("S1", "S2"... "S8") To be used in FC1 as shown in the truth table in Appendix 2.
Q0.6	BOOL	Physical output. To be used in FC1 as shown in the Boolean table in Appendix 2.
Q0.7	BOOL	Physical output. To be used in FC1 as shown in the Boolean table in Appendix 2.
M82.2	BOOL	Marker for the control room. To be used in FC1 as shown in the Boolean table in Appendix 2.
M82.3	BOOL	Marker for the control room. To be used in FC1 as shown in the Boolean table in Appendix 2.
M90	BOOL	Enable/Disable from WinCC '1' = Enabled (ON), '0' = Disabled (OFF).

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Annex 2. Truth table that FC1's program must implement (also note the position of the Enable/Disable switch, which "appears" on the marker M90.0)

I0.0	I0.1	I0.2	I0.3	Q0.6 ja M82.2	Q0.7 ja M82.3
0	0	0	0	0	0
0	0	0	1	0	0
0	0	1	0	0	0
0	0	1	1	0	0
0	1	0	0	0	0
0	1	0	1	0	1
0	1	1	0	0	0
0	1	1	1	0	0
1	0	0	0	0	0
1	0	0	1	0	0
1	0	1	0	1	0
1	0	1	1	0	0
1	1	0	0	0	0
1	1	0	1	0	0
1	1	1	0	0	0
1	1	1	1	0	0